Telco Customer Churn Analysis

Executive Summary | IBM Telco Customer Churn dataset (7,043 customers, 21 attributes) | Python: pandas, matplotlib, seaborn

Objective

The analysis explores customer churn patterns across contract type, payment method, tenure, internet service, add-on services and demographics. The goal is to identify which factors are most strongly associated with higher churn, to guide customer retention strategy.

Overall churn rate: 26.5% (1,869 of 7,043 customers churned; 5,174 stayed).

Data Preparation

- **TotalCharges** was stored as text because 11 rows contained a blank value. All 11 had a tenure of 0 months (new customers not yet billed), so the blanks were set to 0 and the column converted to a numeric type.
- No missing values remained and there were no duplicate customer IDs.
- **SeniorCitizen** was recoded from 0/1 to no/yes for readable, categorical analysis.

Key Insights & Findings

1. Contract Type and Churn

- Customers on **month-to-month** contracts have the highest churn rate at **42.7%**.
- Customers on **one-year** and **two-year** contracts churn at **11.3%** and **2.8%** respectively.
- 88.6% of all churned customers were on month-to-month contracts.

Implication: Longer contract periods are a strong retention lever; customers with extended commitments are far less likely to leave.

2. Payment Method and Churn

- Customers paying by **electronic check** churn at **45.3%**.
- Other methods churn far less: mailed check **19.1%**, bank transfer (automatic) **16.7%**, credit card (automatic) **15.2%**.
- Electronic-check users also have a much shorter median tenure (18 months vs 47–48 months for automatic payments).

Implication: Manual monthly payment, convenience or trust issues may contribute. Encouraging a move to automatic payment methods could reduce churn.

3. Churn by Tenure

- Customers in their **first 12 months** churn at **47.4%**; 61% of customers in their first month churn.
- Churn falls to **25.5%** for 13–36 months of tenure and to **11.9%** beyond 36 months.
- Median tenure of churned customers is 10 months, versus 38 months for retained customers.

Implication: Engaging customers early in their journey, especially within the first year, is critical for retention.

4. Churn by Internet Service Type

- **Fiber optic** customers churn at **41.9%**, more than double **DSL** customers at **19.0%**.
- Customers with no internet service churn at only 7.4%.

Implication: Higher prices, competition or service-quality dissatisfaction may drive fiber churn; customer satisfaction with speed and reliability should be investigated.

5. Add-on Services

- Customers **without Online Security** churn at 41.8% vs 14.6% with it; **without Tech Support** 41.6% vs 15.2% with it.
- Streaming TV and Streaming Movies make little difference (about 30% with vs 34% without).

Implication: Support and security services are associated with stronger retention and are good candidates for bundles.

6. Senior Citizens and Churn

- **Senior citizens** churn at **41.7%**, compared with **23.6%** for non-senior customers.
- Seniors make up 16.2% of the customer base.
- Gender shows no meaningful difference (female 26.9%, male 26.2%).

Implication: Targeted customer service and retention programmes for senior customers may reduce churn in this group.

Visualizations & Data Insights

Count plots, a tenure histogram, a pie chart of overall churn and a stacked percentage bar chart (senior citizens) show the disparity in churn across contract types, payment methods, tenure and services. The table below summarises the churn rate for each key segment.

Factor	Segment	Customers	Churn rate
Overall	All customers	7,043	26.5%
Contract	Month-to-month	3,875	42.7%
	One year	1,473	11.3%
	Two year	1,695	2.8%
Payment method	Electronic check	2,365	45.3%
	Mailed check	1,612	19.1%
	Bank transfer (automatic)	1,544	16.7%
	Credit card (automatic)	1,522	15.2%
Tenure	0–12 months	2,186	47.4%
	13–36 months	1,856	25.5%
	37–72 months	3,001	11.9%
Internet service	Fiber optic	3,096	41.9%
	DSL	2,421	19.0%

Factor	Segment	Customers	Churn rate
	No internet	1,526	7.4%
Senior citizen	Yes	1,142	41.7%
	No	5,901	23.6%

Highest-Risk Profile

Customers who are on a **month-to-month contract**, pay by **electronic check** and use **fiber optic** internet churn at **60.4%** (1,307 customers) — more than double the overall rate. Churned customers also pay more per month (median $79.65 vs $64.43) and account for about 30% of total monthly charges.

Recommendations

- **Promote long-term contracts:** offer incentives for month-to-month customers to commit to one- or two-year contracts.
- **Address payment method concerns:** encourage customers to switch from electronic checks to automatic bank transfer or credit card payments.
- **Customer engagement in early tenure:** focus on onboarding and experience within the first year, when churn is highest.
- **Bundle support services:** promote Online Security and Tech Support, especially to fiber optic customers.
- **Senior citizen retention programmes:** create personalised offers or assistance for senior customers.

Note: these findings are correlational; testing each intervention (e.g. an A/B test) would confirm its effect on churn.